

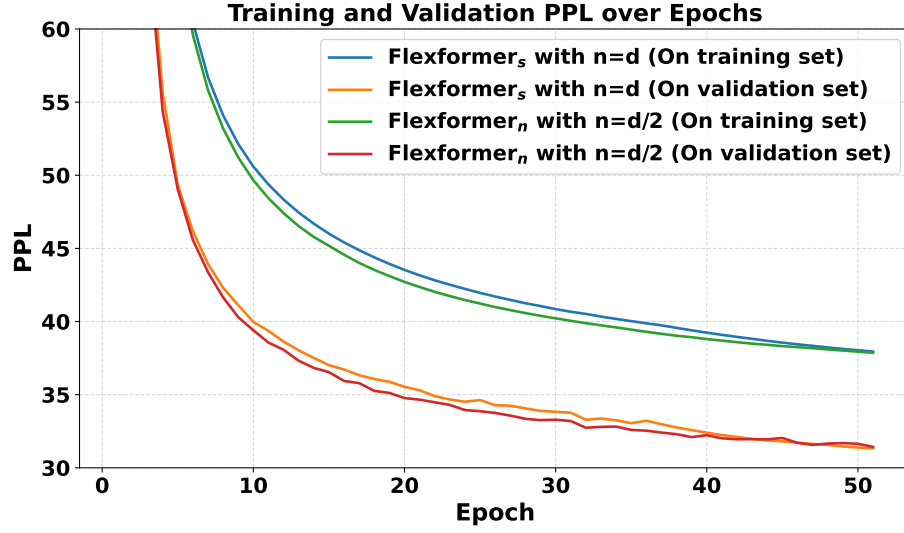


Figure 1: Comparison of the PPL curves on WikiText-103 between Flexformer_s with $n=d$ and Flexformer_n with $n = \frac{d}{2}$, which have the same number of parameters. It can be observed that the two models achieve similar performance, while Flexformer_n converges faster.